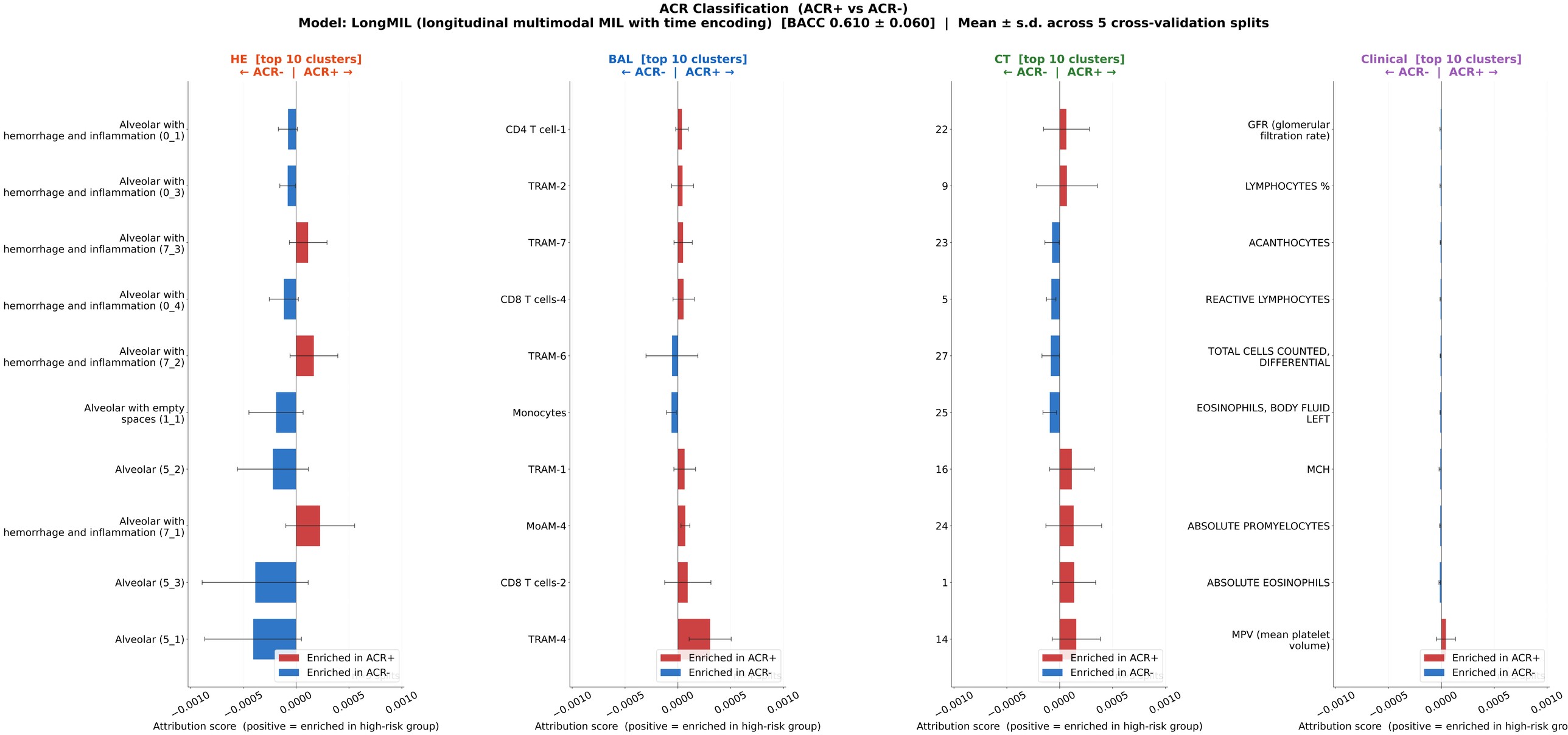
ACR Classification (ACR+ vs ACR-)
Model: LongMIL (longitudinal multimodal MIL with time encoding) [BACC 0.610 ± 0.060] | Mean ± s.d. across 5 cross-validation splits
HE [top 10 clusters]
← ACR- | ACR+ →
Alveolar with hemorrhage and inflammation (0_1)
Alveolar with hemorrhage and inflammation (0_3)
Alveolar with hemorrhage and inflammation (7_3)
Alveolar with hemorrhage and inflammation (0_4)
Alveolar with hemorrhage and inflammation (7_2)
Alveolar with empty spaces (1_1)
Alveolar (5_2)
Alveolar with hemorrhage and inflammation (7_1)
Alveolar (5_3)
Alveolar (5_1)
Enriched in ACR+
Enriched in ACR-
−0.0010
−0.0005
0.0000
0.0005
0.0010
Attribution score (positive = enriched in high-risk group)
BAL [top 10 clusters]
← ACR- | ACR+ →
CD4 T cell-1
TRAM-2
TRAM-7
CD8 T cells-4
TRAM-6
Monocytes
TRAM-1
MoAM-4
CD8 T cells-2
TRAM-4
Enriched in ACR+
Enriched in ACR-
−0.0010
−0.0005
0.0000
0.0005
0.0010
Attribution score (positive = enriched in high-risk group)
CT [top 10 clusters]
← ACR- | ACR+ →
22
9
23
5
27
25
16
24
1
14
Enriched in ACR+
Enriched in ACR-
−0.0010
−0.0005
0.0000
0.0005
0.0010
Attribution score (positive = enriched in high-risk group)
Clinical [top 10 clusters]
← ACR- | ACR+ →
GFR (glomerular filtration rate)
LYMPHOCYTES %
ACANTHOCYTES
REACTIVE LYMPHOCYTES
TOTAL CELLS COUNTED, DIFFERENTIAL
EOSINOPHILS, BODY FLUID LEFT
MCH
ABSOLUTE PROMYELOCYTES
ABSOLUTE EOSINOPHILS
MPV (mean platelet volume)
Enriched in ACR+
Enriched in ACR-
−0.0010
−0.0005
0.0000
0.0005
0.0010
Attribution score (positive = enriched in high-risk gro